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### Failover of cloud-native network functions within node groups for high availability in a wireless telecommunication network

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#### Abstract

Example embodiments are directed towards detecting a failure of a fifth-generation New Radio (5G NR) cellular telecommunication network containerized network function (CNF) running within a dedicated node group of cloud compute instances hosting a plurality of CNFs of a wireless telecommunication service provider or a failure of a specific cloud compute instance hosting the CNF within the node group. In response to the detection of a failure of the CNF within the node group or a failure of the specific cloud compute instance hosting the CNF within the node group, automatically switching to host the CNF on a spare cloud compute instance within the node group dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider.

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## Background/Summary

### TECHNICAL FIELD

(1) The present disclosure relates generally to telecommunication networks, more particularly, to failover of cloud-native network functions (CNFs) in a wireless telecommunication network.

### BRIEF SUMMARY

(2) It is advantageous to provide Fifth Generation (5G) wireless technology in a highly available, flexible manner that avoids reduces network outages and dropped calls experienced by ends users. It is with respect to these and other considerations that the embodiments described herein have been made.

(3) 5G provides a broad range of wireless services delivered to the end user across multiple access platforms and multi-layer networks. 5G is a dynamic, coherent and flexible framework of multiple advanced technologies supporting a variety of applications. 5G utilizes an intelligent architecture, with Radio Access Networks (RANs) not constrained by base station proximity or complex

infrastructure. 5G enables a disaggregated, flexible and virtualized RAN with interfaces creating additional data access points. 5G network functions may be completely software-based and designed as cloud-native, meaning that they're agnostic to the underlying cloud infrastructure, allowing higher deployment, agility and flexibility. With the advent of 5G, industry experts defined how the 5G core (5GC) network should evolve to support the needs of 5G New Radio (NR) and the advanced use cases enabled by it. The 3rd Generation Partnership Project (3GPP) develops protocols and standards for telecommunication technologies including RAN, core transport networks and service capabilities. 3GPP has provided complete system specifications for 5G network architecture which is much more service oriented than previous generations.

(4) Multi-Access Edge Computing (MEC) is an important element of 5G architecture. MEC is an evolution in cloud computing that brings the applications from centralized data centers to the network edge, and therefore closer to the end users and their devices. This essentially creates a shortcut in content delivery between the user and host, and the long network path that once separated them.

(5) This MEC technology is not exclusive to 5G but is certainly important to its efficiency. Characteristics of the MEC include the low latency, high bandwidth and real time access to RAN information that distinguishes 5G architecture from its predecessors. This convergence of the RAN and core networks enables operators to leverage new approaches to network testing and validation. 5G networks based on the 3GPP 5G specifications provide an environment for MEC deployment. The 5G specifications define the enablers for edge computing, allowing MEC and 5G to collaboratively route traffic. In addition to the latency and bandwidth benefits of the MEC architecture, the distribution of computing power is better enables the high volume of connected devices inherent to 5G deployment and the rise of IoT.

(6) The 3rd Generation Partnership Project (3GPP) develops protocols for mobile telecommunications and has developed a standard for 5G. The 5G architecture is based on what is called a Service-Based Architecture (SBA), which implements IT network principles and a cloud-native design approach. In this architecture, each network function (NF) offers one or more services to other NFs via Application Programming Interfaces (API). Network function virtualization (NFV) decouples software from hardware by replacing various network functions such as firewalls, load balancers and routers with virtualized instances running as software. This eliminates the need to invest in many expensive hardware elements and can also accelerate installation times, thereby providing revenue generating services to the customer faster.

(7) NFV enables the 5G infrastructure by virtualizing appliances within the 5G network. This includes the network slicing technology that enables multiple virtual networks to run simultaneously. NFV may address other 5G challenges through virtualized computing, storage, and network resources that are customized based on the applications and customer segments. The concept of NFV extends to the RAN through, for example, network disaggregation promoted by alliances such as O-RAN. This enables flexibility, provides open interfaces and open source development, ultimately to ease the deployment of new features and technology with scale. The O-RAN ALLIANCE objective is to allow multi-vendor deployment with off-the shelf hardware for the purposes of easier and faster inter-operability. Network disaggregation also allows components of the network to be virtualized, providing a means to scale and improve user experience as capacity grows. The benefits of virtualizing components of the RAN provide a means to be more cost effective from a hardware and software viewpoint especially for IoT applications where the number of devices is in the millions.

(8) The 5G New Radio (5G NR) RAN comprises of a set of radio base stations (each known as Next Generation Node B (gNb)) connected to the 5G core (5GC) and to each other. The gNb incorporates three main functional modules: the Centralized Unit (CU), the distributed Unit (DU), and the Radio Unit (RU), which can be deployed in multiple combinations. The primary interface is referred to as the F1 interface between DU and CU and are interoperable across vendors. The CU

may be further disaggregated into the CU user plane (CU-UP) and CU control plane (CU-CP), both of which connect to the DU over F1-U and F1-C interfaces respectively. This 5G RAN architecture is described in 3GPP TS 38.401 V16.8.0 (2021-12). Each network function (NF) is formed by a combination of small pieces of software code called as microservices.

(9) Embodiments, described herein may use containerization to implement such microservices. Containerization is the packaging of software code with just the operating system (OS) libraries and dependencies required to run the code to create a single lightweight executable (a container) that runs consistently on any infrastructure. Software platforms, such as Kubernetes, manage containerized workloads and automate the deployment, scaling, and management of containerized applications. Compared to virtual machines (VMs) containers have relaxed isolation properties to share the Operating System (OS) among the applications. Therefore, containers are considered lightweight. A container has its own file system, share of CPU, memory and process space. As they are decoupled from the underlying infrastructure, they are portable across clouds and OS distributions.

(10) A cluster is made up of nodes that run containerized applications. Each cluster also has a master (control plane) that manages the nodes and pods of the cluster. A node represents a single machine in a cluster, typically either a physical machine or virtual machine that's located either on-premises or hosted by a cloud service provider. Each node hosts groups of one or more containers (which run applications), and the master communicates with nodes about when to create or destroy containers and how to re-route traffic based on new container alignments. The Kubernetes master is the access point (or the control plane) from which administrators and other users interact with the cluster to manage the scheduling and deployment of containers.

(11) A pod is the basic unit of scheduling for applications running on a cluster. The applications are running in containers, and each pod comprises one or more container(s). While pods are able to house multiple containers, one-container-per-pod may also be used. In some situations, containers that are tightly coupled and need to share resources may sit in the same pod. Pods can quickly and easily communicate with one another as if they were running on the same machine. They do still, however, maintain a degree of isolation. Each pod is assigned a unique IP address within the cluster, allowing the application to use ports without conflict.

(12) Pods are designed as relatively ephemeral, disposable entities. When a pod gets created, it is scheduled to run on a node. The pod remains on that node until the process is terminated, the pod object is deleted, the pod is evicted for lack of resources, or the node fails. In Kubernetes, pods are the unit of replication. If an application becomes overly popular and a pod can no longer facilitate the load, Kubernetes can deploy replicas of the pod to the cluster.

(13) Software container orchestration platforms, such as Amazon Elastic Kubernetes Service (Amazon EKS), are services for users to run Kubernetes on the cloud of a cloud computing service provider, such as Amazon Web Services (AWS), without the user needing to install, operate, and maintain their own Kubernetes control plane or nodes. An Amazon EKS cluster comprises of two primary components: the Amazon EKS control plane and Amazon EKS nodes that are registered with the control plane. The Amazon EKS control plane comprises of control plane nodes that run the Kubernetes software and the Kubernetes application programming interface (API) server. The control plane may run in an account managed by AWS or the telecommunication service provider, and the Kubernetes API is exposed via the Amazon EKS endpoint associated with the cluster. Each Amazon EKS cluster control plane is single-tenant and unique, and runs on its own set of Amazon Elastic Compute Cloud (EC2) instances, which provide scalable computing capacity in the Amazon Web Services (AWS) cloud. However, other types of cloud compute instances or virtual machine instances may be used on various other cloud computing provider service platforms. The cluster control plane may be provisioned across multiple Availability Zones (AZs) and fronted by an Elastic Load Balancing Network Load Balancer. Amazon EKS may also provision elastic network interfaces in VPC subnets to provide connectivity from the control plane instances to the nodes.

Amazon EKS nodes may run in an AWS account of the telecommunication service provider and connects to the telecommunication service provider's cluster control plane via the API server endpoint and a certificate file that is created for the cluster.

(14) As disclosed herein, NFs of the 5G NR cellular telecommunication network implemented in respective node groups are useful mechanisms for creating pools of resources in the 5G network that can enforce scheduling requirements. They also provide a utility for shifting workloads around in the 5G network during cluster management and updates. Such NFs of the 5G NR cellular telecommunication network may be hosted on a cloud service provider cloud and referred to herein as cloud-native network functions (CNFs).

(15) Briefly described, embodiments disclosed herein are directed toward detecting a failure of a fifth-generation New Radio (5G NR) cellular telecommunication network containerized network function (CNF) running within a dedicated node group of cloud compute instances hosting a plurality of CNFs of a wireless telecommunication service provider or a failure of a specific cloud compute instance hosting the CNF within the node group. In response to the detection of a failure of the CNF within the node group or a failure of the specific cloud compute instance hosting the CNF within the node group, automatically switching to host the CNF on a spare cloud compute instance within the node group dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider. In an example embodiment, the CNF corresponds to and is implemented by one or more software containers of one or more respective pods for scheduling and execution on one or more respective nodes in a software container orchestration platform that automates deployment, management, and scaling of containerized software applications.

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## Description

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) Non-limiting and non-exhaustive embodiments are described with reference to the following drawings. In the drawings, like reference numerals refer to like parts throughout the various figures unless otherwise specified.

(2) For a better understanding of the present invention, reference will be made to the following Detailed Description, which is to be read in association with the accompanying drawings:

(3) FIG. 1 illustrates a diagram of an example system architecture overview of a system in which failover of cloud-native network functions (CNFs) within node groups for high availability in a wireless telecommunication network may be implemented in accordance with embodiments described herein.

(4) FIG. 2A is a diagram of a cloud-native system for failover of CNFs within node groups for high availability in a wireless telecommunication network in accordance with an embodiment described herein.

(5) FIG. 2B is a diagram of a cloud-native system for failover of cloud-native network functions (CNFs) within node groups for high availability in a wireless telecommunication network in accordance with another embodiment described herein.

(6) FIG. 3 illustrates a logical flow diagram showing an example embodiment of a process for failover of CNFs within node groups for high availability in a wireless telecommunication network in accordance with embodiments described herein.

(7) FIG. 4 illustrates a logical flow diagram showing an example embodiment of a process for failover of 5G wireless network centralized unit control plane (CU-CP) CNFs within a CU-CP node group for high availability in a wireless telecommunication network in accordance with embodiments described herein.

(8) FIG. 5 shows a system diagram that describes an example implementation of a computing

system(s) for implementing embodiments described herein.

## DETAILED DESCRIPTION

(9) The following description, along with the accompanying drawings, sets forth certain specific details in order to provide a thorough understanding of various disclosed embodiments. However, one skilled in the relevant art will recognize that the disclosed embodiments may be practiced in various combinations, without one or more of these specific details, or with other methods, components, devices, materials, etc. In other instances, well-known structures or components that are associated with the environment of the present disclosure, including but not limited to the communication systems and networks, have not been shown or described in order to avoid unnecessarily obscuring descriptions of the embodiments. Additionally, the various embodiments may be methods, systems, media, or devices. Accordingly, the various embodiments may be entirely hardware embodiments, entirely software embodiments, or embodiments combining software and hardware aspects.

(10) Throughout the specification, claims, and drawings, the following terms take the meaning explicitly associated herein, unless the context clearly dictates otherwise. The term “herein” refers to the specification, claims, and drawings associated with the current application. The phrases “in one embodiment,” “in another embodiment,” “in various embodiments,” “in some embodiments,” “in other embodiments,” and other variations thereof refer to one or more features, structures, functions, limitations, or characteristics of the present disclosure, and are not limited to the same or different embodiments unless the context clearly dictates otherwise. As used herein, the term “or” is an inclusive “or” operator, and is equivalent to the phrases “A or B, or both” or “A or B or C, or any combination thereof,” and lists with additional elements are similarly treated. The term “based on” is not exclusive and allows for being based on additional features, functions, aspects, or limitations not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of “a,” “an,” and “the” include singular and plural references.

(11) FIG. 1 illustrates a diagram of an example system architecture overview of a system in which failover of cloud-native network functions (CNFs) within node groups for high availability in a wireless telecommunication network may be implemented in accordance with embodiments described herein.

(12) The system **100** illustrates an example architecture of at least one wireless network of a mobile network operator (MNO) that is operated and/or controlled by the MNO. The system may comprise a 5G wireless cellular telecommunication network including a disaggregated, flexible and virtual RAN with interfaces creating additional data access points and that is not constrained by base station proximity or complex infrastructure. As shown in FIG. 1, a 5G RAN is split into DUs (e.g., DU **104**) that manage scheduling of all the users and a CU **102** that manages the mobility and radio resource control (RRC) state for all the UEs. The RRC is a layer within the 5G NR protocol stack. The CU **102** is hosted within one or more local zones of the cloud computing service provider. For example, local zone 1 (LZ-1) **124**. A local zone is a type of infrastructure deployment of the cloud computing service provider that places compute, storage, database, and/or other select cloud computing services closer to end users of the services and systems hosted by cloud computing service provider (e.g., closer to large population and industry centers). Local zones enable users of the cloud computing service provider cloud to use select cloud computing services, like compute and storage services, closer to more end-users, providing them very low latency access to the applications running locally. Local Zones are also connected to the parent region (e.g., Region A, Region B and/or Region C shown in FIG. 1) via the cloud service provider's redundant and very high bandwidth private network, giving applications running in local zones fast, secure, and seamless access to the rest of the cloud computing services.

(13) As shown in FIG. 1, the radio unit (RU) **106** converts radio signals sent to and from the antenna of base stations **122** into a digital signal for transmission over packet networks. It handles the digital front end (DFE) and the lower physical (PHY) layer, as well as the digital beamforming

functionality.

(14) The DU **104** may sit close to the RU **106** and runs the radio link control (RLC), the Medium Access Control (MAC) sublayer of the 5G NR protocol stack, and parts of the PHY layer. The MAC sublayer interfaces to the RLC sublayer from above and to the PHY layer from below. The MAC sublayer maps information between logical and transport channels. Logical channels are about the type of information carried whereas transport channels are about how such information is carried. This logical node includes a subset of the gNb functions, depending on the functional split option, and its operation is controlled by the CU **102**.

(15) The CU **102** is the centralized unit that runs the RRC and Packet Data Convergence Protocol (PDCP) layers. A gNb may comprise a CU and one DU connected to the CU via Fs-C and Fs-U interfaces for control plane (CP) and user plane (UP) respectively. A CU with multiple DUs will support multiple gNbs. The split architecture enables a 5G network to utilize different distribution of protocol stacks between CU **102** and DU **104** depending on midhaul availability and network design. The CU **102** is a logical node that includes the gNb functions like transfer of user data, mobility control, RAN sharing, positioning, session management etc., with the exception of functions that may be allocated exclusively to the DU **104**. The CU **102** controls the operation of several DUs **104** over the midhaul interface.

(16) As mentioned above, 5G network functionality is split into two functional units: the DU **104**, responsible for real time 5G layer 1 (L1) and 5G layer 2 (L2) scheduling functions, and the CU **102** responsible for non-real time, higher L2 and 5G layer 3 (L3). As shown in FIG. 1, the DU's server and relevant software may be hosted on a cell site **116** itself or can be hosted in an edge cloud (local data center (LDC) **118** or central office) depending on transport availability and fronthaul interface. DUs from multiple cell sites may be pooled and col-located at one LDC **118**. The CU's server and relevant software may be hosted in a regional cloud data center or, as shown in FIG. 1, in a breakout edge data center (B-EDC) **114**. As shown in FIG. 1, the DU **104** may be provisioned to communicate via a pass through edge data center (P-EDC) **108**. The P-EDC **108** may provide a direct circuit fiber connection from the DU directly to the primary physical data center (e.g., B-EDC **114**) hosting the CU **102**. In some embodiments, the LDC **118**, P-EDC **108** and/or the B-EDC **114** may be co-located or in a single location. The CU **102** may be connected to a regional cloud data center (RDC) **110**, which in turn may be connected to a national cloud data center (NDC) **112**. In the example embodiment, the P-EDC **108**, the LDC **118**, the cell site **116** and the RU **106** may all be managed and/or controlled by the mobile network operator and the B-EDC **114**, the RDC **110** and the NDC **112** may all be managed and/or hosted by a cloud computing service provider. In some embodiments, the P-EDC **108**, LDC **118** and cell site **116** may be at a single location or facility (e.g., a colocation data center). In other embodiments, the B-EDC **114**, the P-EDC **108**, the LDC **118** and cell site **116** may be at a single location or facility (e.g., a colocation data center). According to various embodiments, the actual split between DU and RU may be different depending on the specific use-case and implementation.

(17) FIG. 2A is a diagram of a cloud-native system **202** for failover of CNFs within node groups for high availability in a wireless telecommunication network in accordance with an embodiment described herein.

(18) In the present example embodiment of FIG. 2, shown is a cluster **206** of cloud-native network function (CNF) node groups, such as node group **208**, node group **226** and node group **238**, hosted within a local zone **204** of a cloud computing service provider cloud. In an example embodiment, the CNF corresponds to and is implemented by one or more software containers of one or more respective pods for scheduling and execution on one or more respective nodes in a software container orchestration platform that automates deployment, management, and scaling of containerized software applications. The node groups are those of cloud compute instances hosting a plurality of 5G NR centralized unit (CU) CNFs of a wireless telecommunication service provider. For example, the node groups may be those of cloud compute instances hosting the CU **102** within

cloud computing service provider local zone 1 (AZ-1) **124** of FIG. **1**. Each node group is overprovisioned with one or more spare cloud compute instances within the node group in case any of the CNF instances or eight cloud compute instances of the node group fails. The spare cloud compute instances are dedicated to the wireless telecommunication service provider and pre-provisioned to host CU CNFs of a same type as that hosted by the node group.

(19) For example, CU control plane (CU-CP) node group **208** is generated to include eight cloud compute instances (cloud compute instance 1 **210**, cloud compute instance 2 **215** to cloud compute instance 8 **216**) dedicated to the wireless telecommunication service provider, with each cloud compute instance of the eight cloud compute instances dedicated to hosting three respective CU-CP CNF instances. For example, cloud compute instance 1 **210** hosts the CU-CP CNF instances CU-CP-1 **212** to CU-CP-3 **214**. CU-CP node group **208** is also overprovisioned by being generated to also include two spare cloud compute instances for the CU-CP node group **208** (spare cloud compute instance **218** and spare cloud compute instance **220**) dedicated to the wireless telecommunication service provider. Each spare cloud compute instance for the CU-CP node group **208** (e.g., spare cloud compute instance **218**) is pre-provisioned to host three CU-CP CNF instances in case any CU-CP CNF instances of the CU-CP node group **208** fails or any of the eight cloud compute instances (cloud compute instance 1 **210**, cloud compute instance 2 **215** to cloud compute instance 8 **216**) fails.

(20) A control plane of the cluster **206** may detect such failures and control automatically switching over to the spare cloud compute instances. In an example embodiment, the cloud computing service provider is Amazon Web Services (AWS), the cloud compute instances are Amazon Elastic Compute Cloud (EC2) instances and the software container orchestration platform is the Amazon Elastic Kubernetes Service (Amazon EKS). For example, Kubernetes configurations may be used to detect and relocate impacted pods in the event of failure to another EC2 within the node group. In the present example, a single EKS cluster (e.g., cluster 1 **206**) will manage 24 CU pairs (24 CU-CPs in CU-CP Node Group **208**, 24 CU user plane CNF instances (CU-UPs) across CU-CP Node Group 1 **226** and CU-CP Node Group 2 **238**) and management nodes in Node Group **239**. In particular, CU-CP Node Group 1 **226** includes 12 CU-UP CNF instances in four EC2 instances, which include EC2 instance 1 **228** to EC2 instance 4 **230**. This provides the ability to support up to 10 Gbps traffic across 12 CU-UPs. CU-CP Node Group 1 **226** also includes a spare EC2 instance **232**. However, other cloud computing service providers, cloud compute instances and software container orchestration platforms may be used.

(21) Also, in the present embodiment, all CU-CPs within an EKS cluster (e.g., cluster 1 **206**) will be part of the same node group (e.g., CU-CP Node Group **208**) and are mapped to a single virtual router (V-Router) pair **222**. Control elements (such as, for example, Mavenir control elements software deployment as a service (SDaaS), Mavenir Telco Cloud Integration Layer (MTCIL), and Cloud Range Data Layer (CRDL)) are in a separate Node Group **239**, but mapped to the same V-Router pair **222** managing CU-CP Node Group **208**. Also, CU-CUP Node Group 1 **226** is mapped to a single V-Router pair **234** and CU-CUP Node Group 2 **238** (which also includes a spare EC2 instance) is mapped to a single V-Router pair **235**. In an example embodiment, the overprovisioning the dedicated node group (e.g., CU-CP Node group **208**) with one or more spare cloud compute instances includes overprovisioning the dedicated node group such that 20% of the cloud compute instances within the node group are spare cloud compute instances.

(22) FIG. **2B** is a diagram of a cloud-native system **239** for failover of cloud-native network functions (CNFs) within node groups for high availability in a wireless telecommunication network in accordance with another embodiment described herein.

(23) The system **239** of FIG. **2B** shares commonalities with the system **202** of FIG. **2A**, except each node group is over provisioned with additional/active cloud compute instances (e.g., EC2(s)) to account for cloud compute instance failures for both CU-UP and CU-CP. For example, this provides the ability to support up to 10 Gbps traffic across 8 CU-UPs. For example, CU Node



Group 1 **240** is generated to include x cloud compute instances (cloud compute instance 1 **242** to cloud compute instance x-1 **249**) for CU CNFs dedicated to the wireless telecommunication service provider, with each cloud compute instance of the x cloud compute instances dedicated to hosting two respective CU CNF instances, such as CU-1 **246** and CU-2 **247**. Furthermore, each node group (CU Node Group 1 **240**, CU Node Group 2 **254** and CU Node Group 3 **256** is mapped to a respective pair of V-Routers to increase high availability of telecommunication services (e.g., in case one V-Router of the pair fails). For example, CU Node Group 1 **240** is mapped to V-Router pair **250**.

(24) FIG. **3** illustrates a logical flow diagram showing an example embodiment of a process **300** for failover of CNFs within node groups for high availability in a wireless telecommunication network in accordance with embodiments described herein.

(25) At **302** the system **100**, detects a failure of a fifth-generation New Radio (5G NR) cellular telecommunication network cloud-native network function (CNF) running within a dedicated node group of cloud compute instances hosting a plurality of CNFs of a wireless telecommunication service provider or a failure of a specific cloud compute instance hosting the CNF within the node group. The CNF may correspond to and be implemented by one or more software containers of one or more respective pods for scheduling and execution on one or more respective nodes in a software container orchestration platform that automates deployment, management, and scaling of containerized software applications. The dedicated node group may be overprovisioned with one or more spare cloud compute instances within the node group. The spare cloud compute instances are dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider. For example, the CNF is a 5G wireless network centralized unit (CU) network function, a 5G wireless network CU user plane (CU-UP) network function, or a 5G wireless network CU control plane (CU-CP) network function. In an example embodiment, the overprovisioning the dedicated node group with one or more spare cloud compute instances includes overprovisioning the dedicated node group such that 20% of the cloud compute instances within the node group are spare cloud compute instances.

(26) At **304**, the system **100**, in response to the detection of a failure of the CNF within the node group or a failure of the specific cloud compute instance hosting the CNF within the node group, automatically switching to host the CNF on a spare cloud compute instance within the node group dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider.

(27) FIG. **4** illustrates a logical flow diagram showing an example embodiment of a process **400** for failover of 5G wireless network centralized unit control plane (CU-CP) CNFs within a CU-CP node group for high availability in a wireless telecommunication network in accordance with embodiments described herein.

(28) At **402**, the system **100**, generates a CU-CP node group including eight cloud compute instances dedicated to the wireless telecommunication service provider, with each cloud compute instance of the eight cloud compute instances dedicated to hosting three respective CU-CP CNF instances.

(29) At **404**, the system **100**, generates two spare cloud compute instances for the CU-CP node group dedicated to the wireless telecommunication service provider and pre-provisioned to host three CU-CP CNF instances in case any CU-CP CNF instances of the CU-CP node group fails or any of the eight cloud compute instances fails.

(30) At **406**, the system **100**, detects failure of one of the eight cloud compute instances dedicated to hosting CU-CP CNF instances.

(31) At **408**, the system **100**, in response to detecting failure of one of the eight cloud compute instances, automatically switching to host the three respective CU-CP CNF instances of the failed cloud compute instance on one of the two spare cloud compute instances for the CU-CP node group dedicated to the wireless telecommunication service provider.

(32) FIG. 5 shows a system diagram that describes an example implementation of a host computing system or systems **501** for implementing embodiments described herein.

(33) The functionality described herein for failover of cloud-native network functions (CNFs) within node groups for high availability in a wireless telecommunication network can be implemented either on dedicated hardware, as a software instance running on dedicated hardware, or as a virtualized function instantiated on an appropriate platform, e.g., a cloud infrastructure. In some embodiments, such functionality may be completely software-based and designed as cloud-native, meaning that they are agnostic to the underlying cloud infrastructure, allowing higher deployment agility and flexibility. However, FIG. 5 illustrates an example of underlying hardware on which such software and functionality may be hosted and/or implemented.

(34) In particular, shown is example host computer system(s) **501**. For example, such computer system(s) **501** may represent one or more of those in various data centers, base stations and cell sites shown and/or described herein that are, or that host or implement the functions of: routers, components, microservices, nodes, node groups, control planes, clusters, virtual machines, NFs, and other aspects described herein for failover of cloud-native network functions (CNFs) within node groups for high availability in a wireless telecommunication network. In some embodiments, one or more special-purpose computing systems may be used to implement the functionality described herein. Accordingly, various embodiments described herein may be implemented in software, hardware, firmware, or in some combination thereof. Host computer system(s) **501** may include memory **502**, one or more central processing units (CPUs) **514**, I/O interfaces **518**, other computer-readable media **520**, and network connections **522**.

(35) Memory **502** may include one or more various types of non-volatile and/or volatile storage technologies. Examples of memory **502** may include, but are not limited to, flash memory, hard disk drives, optical drives, solid-state drives, various types of random access memory (RAM), various types of read-only memory (ROM), neural networks, other computer-readable storage media (also referred to as processor-readable storage media), or the like, or any combination thereof. Memory **502** may be utilized to store information, including computer-readable instructions that are utilized by CPU **514** to perform actions, including those of embodiments described herein.

(36) Memory **502** may have stored thereon control module(s) **1804**. The control module(s) **1804** may be configured to implement and/or perform some or all of the functions of the systems, components and modules described herein failover of cloud-native network functions (CNFs) within node groups for high availability in a wireless telecommunication network. Memory **502** may also store other programs and data **510**, which may include rules, databases, application programming interfaces (APIs), software containers, nodes, pods, clusters, node groups, control planes, software defined data centers (SDDCs), microservices, virtualized environments, software platforms, cloud computing service software, network management software, network orchestrator software, network functions (NF), artificial intelligence (AI) or machine learning (ML) programs or models to perform the functionality described herein, user interfaces, operating systems, other network management functions, other NFs, etc.

(37) Network connections **522** are configured to communicate with other computing devices to facilitate the functionality described herein. In various embodiments, the network connections **522** include transmitters and receivers (not illustrated), cellular telecommunication network equipment and interfaces, and/or other computer network equipment and interfaces to send and receive data as described herein, such as to send and receive instructions, commands and data to implement the processes described herein. I/O interfaces **518** may include a video interfaces, other data input or output interfaces, or the like. Other computer-readable media **520** may include other types of stationary or removable computer-readable media, such as removable flash drives, external hard drives, or the like.

(38) The various embodiments described above can be combined to provide further embodiments.

These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

## Claims

1. A method for failover of cloud-native network functions (CNFs) in a wireless telecommunication network, the method comprising: detecting a failure of a fifth-generation New Radio (5G NR) cellular telecommunication network cloud-native network function (CNF) running within a dedicated node group of cloud compute instances hosting a plurality of CNFs of a wireless telecommunication service provider or a failure of a specific cloud compute instance hosting the CNF within the node group; and in response to the detection of a failure of the CNF within the node group or a failure of the specific cloud compute instance hosting the CNF within the node group, automatically switching to host the CNF on a spare cloud compute instance within the node group dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider.
2. The method of claim 1 wherein the CNF corresponds to and is implemented by one or more software containers of one or more respective pods for scheduling and execution on one or more respective nodes in a software container orchestration platform that automates deployment, management, and scaling of containerized software applications.
3. The method of claim 2 wherein the CNF is a 5G wireless network centralized unit (CU) network function, a 5G wireless network CU user plane (CU-UP) network function, or a 5G wireless network CU control plane (CU-CP) network function.
4. The method of claim 3, further comprising: generating a CU-CP node group including eight cloud compute instances dedicated to the wireless telecommunication service provider, with each cloud compute instance of the eight cloud compute instances dedicated to hosting three respective CU-CP CNF instances; and generating two spare cloud compute instances for the CU-CP node group dedicated to the wireless telecommunication service provider and pre-provisioned to host three CU-CP CNF instances in case any CU-CP CNF instances of the CU-CP node group fails or any of the eight cloud compute instances fails.
5. The method of claim 4, further comprising: detecting failure of one of the eight cloud compute instances dedicated to hosting CU-CP CNF instances; and in response to detecting failure of one of the eight cloud compute instances, automatically switching to host the three respective CU-CP CNF instances of the failed cloud compute instance on one of the two spare cloud compute instances for the CU-CP node group dedicated to the wireless telecommunication service provider.
6. The method of claim 1, further comprising: overprovisioning the dedicated node group with one or more spare cloud compute instances within the node group, the one or more spare cloud compute instances dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider.
7. The method of claim 6 wherein the overprovisioning the dedicated node group with one or more spare cloud compute instances includes overprovisioning the dedicated node group such that 20% of the cloud compute instances within the node group are spare cloud compute instances.
8. A system for implementing telecommunication network functions comprising: at least one memory that stores computer executable instructions; and at least one processor that executes the computer executable instructions to cause actions to be performed, the actions including: detecting a failure of a fifth-generation New Radio (5G NR) cellular telecommunication network cloud-native network function (CNF) running within a dedicated node group of cloud compute instances

hosting a plurality of CNFs of a wireless telecommunication service provider or a failure of a specific cloud compute instance hosting the CNF within the node group; and in response to the detection of a failure of the CNF within the node group or a failure of the specific cloud compute instance hosting the CNF within the node group, automatically switching to host the CNF on a spare cloud compute instance within the node group dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider.

9. The system of claim 8 wherein the CNF corresponds to and is implemented by one or more software containers of one or more respective pods for scheduling and execution on one or more respective nodes in a software container orchestration platform that automates deployment, management, and scaling of containerized software applications.

10. The system of claim 9 wherein the CNF is a 5G wireless network centralized unit (CU) network function, a 5G wireless network CU user plane (CU-UP) network function, or a 5G wireless network CU control plane (CU-CP) network function.

11. The system of claim 10, wherein at least one processor executes the computer executable instructions to cause further actions to be performed, the further actions including: generating a CU-CP node group including eight cloud compute instances dedicated to the wireless telecommunication service provider, with each cloud compute instance of the eight cloud compute instances dedicated to hosting three respective CU-CP CNF instances; and generating two spare cloud compute instances for the CU-CP node group dedicated to the wireless telecommunication service provider and pre-provisioned to host three CU-CP CNF instances in case any CU-CP CNF instances of the CU-CP node group fails or any of the eight cloud compute instances fails.

12. The system of claim 11, wherein at least one processor executes the computer executable instructions to cause further actions to be performed, the further actions including: detecting failure of one of the eight cloud compute instances dedicated to hosting CU-CP CNF instances; and in response to detecting failure of one of the eight cloud compute instances, automatically switching to host the three respective CU-CP CNF instances of the failed cloud compute instance on one of the two spare cloud compute instances for the CU-CP node group dedicated to the wireless telecommunication service provider.

13. The system of claim 8, wherein at least one processor executes the computer executable instructions to cause further actions to be performed, the further actions including: overprovisioning the dedicated node group with one or more spare cloud compute instances within the node group, the one or more spare cloud compute instances dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider.

14. The system of claim 13 wherein the overprovisioning the dedicated node group with one or more spare cloud compute instances includes overprovisioning the dedicated node group such that 20% of the cloud compute instances within the node group are spare cloud compute instances.

15. A non-transitory computer-readable storage medium having computer-executable instructions stored thereon that, when executed by at least one processor, cause the at least one processor to cause actions to be performed, the actions including: detecting a failure of a fifth-generation New Radio (5G NR) cellular telecommunication network cloud-native network function (CNF) running within a dedicated node group of cloud compute instances hosting a plurality of CNFs of a wireless telecommunication service provider or a failure of a specific cloud compute instance hosting the CNF within the node group; and in response to the detection of a failure of the CNF within the node group or a failure of the specific cloud compute instance hosting the CNF within the node group, automatically switching to host the CNF on a spare cloud compute instance within the node group dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider.

16. The non-transitory computer-readable storage medium of claim 15 wherein the CNF

corresponds to and is implemented by one or more software containers of one or more respective pods for scheduling and execution on one or more respective nodes in a software container orchestration platform that automates deployment, management, and scaling of containerized software applications.

17. The non-transitory computer-readable storage medium of claim 16 wherein the CNF is a 5G wireless network centralized unit (CU) network function, a 5G wireless network CU user plane (CU-UP) network function, or a 5G wireless network CU control plane (CU-CP) network function.

18. The non-transitory computer-readable storage medium of claim 17, wherein the computer-executable instructions, when executed by the at least one processor, cause the at least one processor to cause further actions to be performed, the further actions including: generating a CU-CP node group including eight cloud compute instances dedicated to the wireless telecommunication service provider, with each cloud compute instance of the eight cloud compute instances dedicated to hosting three respective CU-CP CNF instances; and generating two spare cloud compute instances for the CU-CP node group dedicated to the wireless telecommunication service provider and pre-provisioned to host three CU-CP CNF instances in case any CU-CP CNF instances of the CU-CP node group fails or any of the eight cloud compute instances fails.

19. The non-transitory computer-readable storage medium of claim 18, wherein the computer-executable instructions, when executed by the at least one processor, cause the at least one processor to cause further actions to be performed, the further actions including: detecting failure of one of the eight cloud compute instances dedicated to hosting CU-CP CNF instances; and in response to detecting failure of one of the eight cloud compute instances, automatically switching to host the three respective CU-CP CNF instances of the failed cloud compute instance on one of the two spare cloud compute instances for the CU-CP node group dedicated to the wireless telecommunication service provider.

20. The non-transitory computer-readable storage medium of claim 15, wherein the computer-executable instructions, when executed by the at least one processor, cause the at least one processor to cause further actions to be performed, the further actions including: overprovisioning the dedicated node group with one or more spare cloud compute instances within the node group, the one or more spare cloud compute instances dedicated to the wireless telecommunication service provider and pre-provisioned to host CNFs of a same type as the CNF for the wireless telecommunication service provider.

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